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RESEARCH ARTICLE

An Improved YOLOv7 Tiny Algorithm for Vehicle and Pedestrian Detection with Occlusion in Autonomous Driving

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Abstract — Future transportation is advancing in the direction of intelligent transportation systems, where an essential part is vehicle and pedestrian detection. Due to the complex urban traffic environment, vehicles and pedestrians in road monitoring have different forms of occlusion problems, resulting in the missed detection of objects. We design an improved you only look once version 7 (YOLOv7) tiny algorithm for vehicle and pedestrian detection under occlusion, with the following four main improvements. In order to locate the object more accurately, 1 × 1 convolution and identity connection are added to the 3 × 3 convolution, and convolution reparameterization is used to enhance the inference speed of the network model. In view of the complex road background and more interference, the coordinate attention was added to the connection part of backbone and neck to enhance the network's capacity to detect the object and lessen interference from other targets. At the same time, before being sent to the detection head, global attention mechanism is added to improve the accuracy of model detection by capturing three-dimensional features. Considering the issue of imbalanced training samples, we propose focal complete intersection over union (CIOU) loss instead of CIOU loss to become the bounding box regression loss, so that the regression process attention to high-quality anchor boxes. Experiments show that the improved YOLOv7 tiny algorithm achieves 82.2% map @ 0.5 in pattern analysis, statistical modelling and computational learning visual object classes dataset, which is 2.8% higher than before the improvement. The performance of map @ 0.5:0.95 is 5.2% better than the previous improvement. The proposed improved algorithm can availably to detect partial occlusion objects.
Keywords — Intelligent transportation system, Vehicle and pedestrian detection, Attention mechanism, Loss function.
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I. Introduction

With the continuous acceleration of urbanization, vehicles have become a necessity of life, the demand for vehicles has also increased, and the traffic congestion in cities has become more and more serious. Traffic congestion not only brings great challenges to urban transportation, but also causes significant economic losses, and the economic losses caused by traffic congestion are inestimable. This problem exists not only in China, but all over the world. Traditional traffic monitoring methods have been unable to address the demands for traffic security and efficiency. Therefore, the intelligent transportation system (ITS) comes into being, which can be used for the detection of vehicles, so as to realize traffic monitoring and management, reduce transportation congestion and boost travel efficiency.

Intelligent transportation system is a key component of the smart city, which integrates computer vision, machine learning, sensor technology and other technologies. It can be used to realize traffic monitoring, traffic flow analysis, traffic accident prediction and other applications. Autonomous driving is one of the cores of intelligent transportation systems, which is able to sense the surrounding environment through sensors and advanced algorithms. As shown in Figure 1, the autonomous vehicle detects vehicles and pedestrians on the road through

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a camera on the roof. By installing different radars in front and rear, the distance between the vehicle and various objects is measured, and the detection results are communicated to the ITS server. Through the perception and fusion of information, the autonomous decision-making of the vehicle is realized. Behind the autonomous driving technology is the support of object detection technology. Object detection technology can identify vehicles, pedestrians, traffic signs and other targets through computer vision and machine learning methods, and provide necessary environmental information for autonomous driving systems.

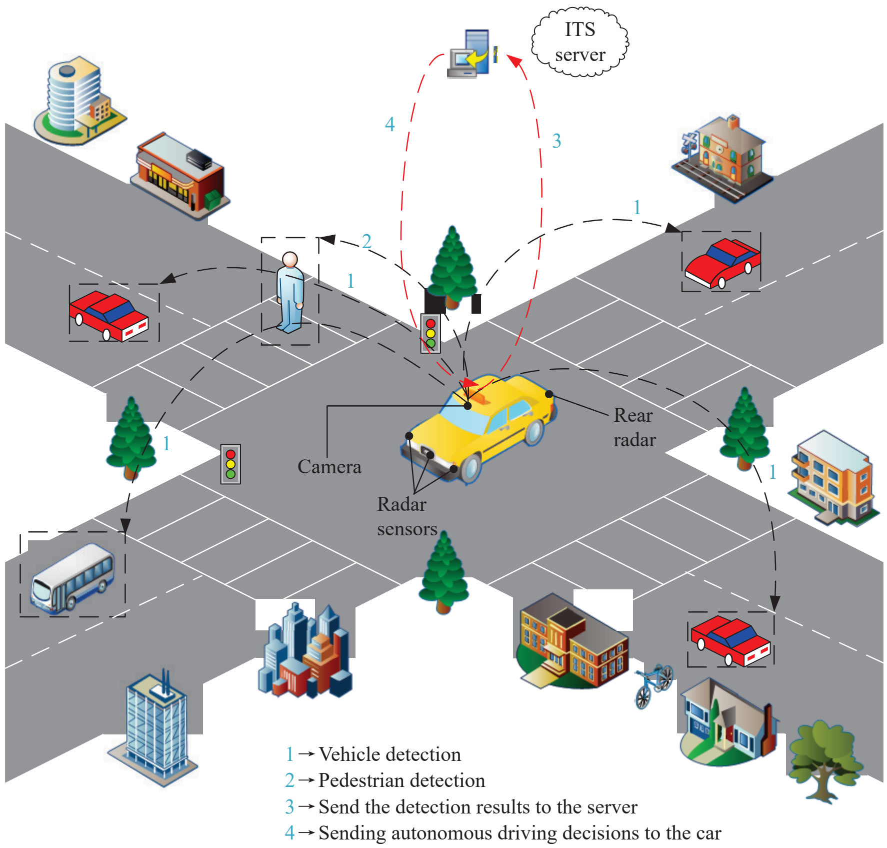

Figure 1 Diagram of the Autonomous driving vehicle.

Object detection is a crucial component of automatic driving, which provides necessary support for the development of automatic driving and has broad development prospects. Since the deep learning algorithm was introduced, detection of objects technology founded on deep learning has entered the mainstream. It can extract vital information from a vast quantity of data, disregard secondary and useless information, and continuously enhance the precision and actual time performance of detection, making the deployment of automated driving and intelligent transportation systems feasible.

Significant progress has been made in vehicle detection in recent years, but there are still some research gaps that need to be addressed. When vehicles are blocked by other objects, such as buildings or other vehicles, occlusion occurs, and it is difficult to detect vehicles and pedestrians in time, which poses a great threat to traffic safety problems, seriously affects traffic efficiency, and even causes serious traffic accidents. In this context, developing effective object detection techniques to deal with occlusion and accurately detect vehicles and pedestrians is an ongoing research challenge. Therefore, developing efficient vehicle and pedestrian detection technologies for occlusion problems can find potential traffic hazards in time, reduce the occurrence of traffic accidents, and ensure the safety of pedestrians and vehicles. Due to the occlusion problem, the existing object detection methods are easy to miss, which has somewhat impacted the accuracy of the algorithm. Therefore, this paper designs a vehicle detection with occlusion for autonomous driving. Considering that the object detection model in autonomous driving requires a network model with less memory and lower complexity, YOLOv7 tiny [1] is selected for improvement.

To solve the above problems, this paper proposes an improved you only look once version 7 (YOLOv7) tiny algorithm for vehicle and pedestrian detection under occlusion, and the main contributions are as follows.

- We design to add a 1×1 convolution and an identity connection to the 3×3 convolution in the YOLOv7 tiny network. In the image under occlusion conditions, more convolutional layers are needed to enrich the features extracted by the network so as to better deal with the object localization problem. In order to improve the inference speed of the model in automatic driving, the reparameterization technology in the inference stage fused the multi-branch convolution structure into a single convolution structure to realize the decoupling of training and prediction. By changing the topology of the network, a faster and stronger vehicle and Pedestrian detection algorithm is realized.

- We design YOLOv7 tiny network to add three coordinate attention (CA) modules to the connection part sent from the backbone to the head. When the vehicle and pedestrian to be detected is occluded, the network model is required to find the region of interest more quickly and accurately, so as to better locate the object, enhance the robustness of the model, and improve the detection ability of the model for occluded objects.
- We propose to add three global attention mechanism (GAM) modules before sending to the last three detection heads. This module captures and effectively utilizes important features by amplifying global interactions with three-dimensional features to reduce information loss. It can better detect the occluded object and improve the accuracy of detection.
- We propose a new regression loss function, focal complete intersection over union (CIOU) loss, to replace CIOU loss. The traditional loss function will have the problem of inaccurate positioning because the image is occluded. Focal CIOU is used to focus the regression process on high-quality anchor boxes so as to better deal with the object location problem and improve the accuracy of object detection.

The rest of this paper is organized as follows. We describe related work on object detection in Section II. In Section III, the improved YOLOv7 tiny model and theoretical basis are introduced. In Section IV, experiments are conducted on publicly available datasets. Finally, the conclusions of this work are provided in Section VI.

II. Related Works

Vehicle and pedestrian detection, which is usually separated into one stage detectors and two stage detectors, is a crucial task in the area of automated driving. The two stage algorithm means that the object detection is divided into two processes: First, the candidate bounding box in the image are extracted, and then the candidate bounding box are classified and position regressed. It usually uses the region proposal network (RPN) to build candidate bounding boxes, and then uses the convolutional neural network (CNN) for classification and position regression. The one-stage algorithm is to extract object locations and categories directly from the image, and it usually uses CNN for feature extraction and classification. The comparison of the related algorithms is displayed in Table 1.

Table 1 Comparison of algorithms

Network	Advantage	Disadvantage
Fast R-CNN [2]	The efficiency is improved by combining object detection and classification into a single network.	The training process requires multiple steps, including selection of candidate regions, feature extraction, and classification, which is complex.
Faster R-CNN [3]	The combination of object detection and candidate region generation reduces the training process of multiple steps and improves the speed and accuracy.	Training and inference in large-scale datasets are still slow.
Mask R-CNN [4]	The ability to use instance segmentation is added to enable detection and segment targets in the image simultaneously.	Greater computational resources are required.
Cascade R-CNN [5]	The cascade structure is introduced to enhance the accuracy of detection through multi-stage cascade detectors.	Greater computational resources are required.
SSD [6]	Multi-scale object detection is achieved by predicting the object's classification and bounding box on the feature maps of different levels.	The detection effect for small targets is poor, and it is easy to produce false detections.
RetinaNet [7]	In the one stage detector, the issue of unbalanced samples that are positive and negative is resolved.	The computation is large and the training time is long.
YOLOv3 [8]	Multi-scale prediction is introduced by performing object detection on feature maps at different levels.	The detection effect for small targets is poor, and it is easy to produce false detections.
YOLOv4 [9]	A larger network structure and more technological innovations are used to boost both the precision and speed of detection.	The computation is large and the training time is long.
YOLOv7 [1]	To increase the model's robustness, a deeper backbone network and multi-scale feature fusion are introduced.	It is easy to produce missed detection in occlusion and blur.
Improved YOLOv5 [10]	An improved feature pyramid model AF-FPN is proposed to improve the accuracy of multi-scale detection while maintaining real-time detection.	It is easy to produce missed detection in terms of motion blur.

1. Two stage detector

A typical two stage object detection algorithm is region based convolutional neural network (R-CNN) [11]. It starts with generating a series of candidate bounding boxes using the selective search (SS) [12] method, and then extracts features for each candidate bounding box, and uses CNN for feature extraction to extract features by performing forward propagation on each candidate bounding box. Fast R-CNN [2] mainly solves the speed problem of R-CNN. It shares the feature extraction process of the whole image. Features are extracted by convolution operations over the entire image and Region of interest (RoI) pooling is used to perform feature extraction for each candidate bounding box. Faster R-CNN [3] introduces the RPN, which is based on fast R-CNN to generate candidate bounding boxes. The window is slid over the feature map and for each window the bounding box and confidence score of the candidate bounding box is generated. A faster object detection speed is achieved. Mask R-CNN [4] is based on faster R-CNN and adds a fully convolutional network branch to generate pixel-level segmentation masks within each candidate bounding box. In this way, not only can the location of the object be located, but also its precise shape may be segmented. Cascade R-CNN [5] is a concatenated object detection algorithm that gradually screens candidate bounding boxes by cascading multiple detectors to increase object detection accuracy and lower false detection rates. However, the speed of the two stage detector is slow and does not meet the requirements of real-time object detection in autonomous driving, the speed of the one stage detector is faster. At the same time, the accuracy of the one stage detector has reached the same level as that of the two stage detector. So the one stage object detector is selected for improvement in this paper.

2. One stage detector

OverFeat [13] is an earlier end-to-end CNN model that extracts candidate bounding boxes from an image using a sliding window, then classifies and positions the candidate bounding boxes using convolutional layers and fully connected layers. However, due to the large amount of calculation in the sliding window, the efficiency is low. single shot multibox detector (SSD) [6] is a detection algorithm based on fully convolutional neural network. Multi-scale object detection is achieved by predicting the object category and bounding box on different levels of feature maps, but the detection effect is poor for small objects, which is prone to false detection. RetinaNet [7] addresses the class imbalance issue by extracting multi-scale features and predicting the class and bounding box of the object at each scale using a feature pyramid network (FPN) [14]. YOLO [15] splits the image into multiple grids and forecasts the bounding box and the corresponding class probabilities on each grid. However, due to the restrictions of coarse grid division and bounding box prediction, there are certain problems in the accuracy of small object detection and localization. YOLOv2 [16] uses finer grid division and introduces anchor boxes to enhance the detection capability of targets of different scales. The Darknet-19 network structure and multi-scale training strategy are also used to improve the accuracy and generalization ability of detection. YOLOv3 [8] introduces multi-scale prediction, which enhances the detection capability of targets at different scales by performing object detection on feature maps at different levels. In addition, in order to enhance the precision and stability of detection, a deeper Darknet-53 network structure and more anchor boxes are being used. YOLOv4 [9] adopts a larger network structure and more technical innovations, including cross stage paritial network (CSP-Darknet53), path aggregation network (PANet) [17], etc., to enhance the precision and speed of detection. YOLOv5 adopts some new network structures, training strategies or technical innovations to improve the detection performance. In order to improve the accuracy of multi-scale detection while maintaining real-time detection, AF-FPN [10] is proposed to replace the original feature pyramid network in YOLOv5, which reduces the loss of information and enhances the expression ability of the feature pyramid. Similar to YOLOv5, YOLOv7 [1] adopts improvement strategies such as deeper backbone network and multi-scale feature fusion, and mainly focuses on the robustness and performance of the improved model. In the context of autonomous driving, despite the improved accuracy of these models, there is still room for improvement in the detection accuracy in terms of object occlusion. At present, the YOLO algorithm has become the mainstream object detection algorithm, which has higher accuracy and speed and has been widely used. YOLOv7 has higher robustness to objects in different scenes by introducing new training techniques. Therefore, this paper chooses the YOLOv7 algorithm for improvement.

3. Vehicle and pedestrian detection under Occlusion

HybridNet [18] adopts a hybrid network structure that combines one stage and two stage detection of objects algorithms. Firstly, YOLOv3 algorithm is used for preliminary vehicle detection to quickly obtain candidate bounding box vehicle frames. It then uses the faster R-CNN algorithm to further refine and classify these candidate bounding boxes to enhance the precision of detection. O-YOLO-v2 [19] resolves the issue of gradient disappearance or discretization by adding convolutional layers and residual modules at different positions of YOLOv2. The network's low-level and high-level characteristics are merged to increase the precision of small vehicle detection. YOLOv7-RAR [20] improves the detection accuracy in urban roads with crowded vehicles by reconstructing the backbone of YOLOv7, adding an attention mechanism and using Gaussian label allocation. Multi-modal attention fusion (MAF)-YOLO [21] proposed an actual-time pedestrian detection algorithm based on multi-modal fusion YOLO, which improved the efficiency of nighttime pedestrian detection through component multi-modal feature extraction module. TOD-YOLOv7 [22] improves the accuracy of occlusion caused by different postures between people by reconstructing the YOLOv7 algorithm, including a small object detection layer and an attention mechanism, and using the recursive gated convolution (g^n Conv, where n denotes the order spatial) [23] module for interaction. Most of the current research is aimed at vehicle and pedestrian detection, and there are still many challenges in the research under occlusion conditions. Therefore, this paper chooses to detect vehicles and pedestrians under occlusion conditions.

III. Methods

1. Convolutional reparameterization

YOLOv7 is divided into three models: YOLOv7, YOLOv7 tiny, and YOLOv7-W6 depending on the GPU. This article makes use of YOLOv7 tiny. Additionally, the models known as YOLOv7-X, YOLOv7-E6, and YOLOv7-D6 were created utilizing composite model scaling [24] for YOLOv7 and YOLOv7-W6 for different service requirements. Expand-efficient layer aggregation networks (E-ELAN) [25] is applied to YOLOv7-E6 to produce YOLOv7-E6E. The three components of YOLOv7 are input, backbone, and head. The input consists of Mosaic [26], MixUp [27], etc. The backbone is made up of ELAN and MP modules; ELAN improves network learning and convergence by controlling the shortest and longest gradient paths, and max pool (MP) implements downsampling by max pooling. The head consists of the spatial pyramid pooling cross stage paritial network (SPPCSP) and ELAN. The SPPCSP module enhances the model performance by enhancing the feature representation ability.

In Algorithm 1, after inputting the dataset, the labeled dataset, and the hyperparameters, Firstly, the model structure is defined, including the convolutional layer, pooling layer, fully connected layer, etc. (Lines 1–3). Secondly, the bounding box loss, class loss, and confidence loss are calculated using the loss function. (Line 4). We can then iterate through training (Lines 5–11), first propagating forward, calculating the loss, and then propagating back. After training, object detection is performed (Lines 14–17), and non-maximum suppression (NMS) is used to filter out anchor boxes; finally, the model is tested (Lines 18–21), the bounding boxes are plotted and the labels of the categories are displayed.

Algorithm 1 YOLOv7 detection process

Input: Dataset, labeles, hyperparameters.

Output: Detection results, bounding boxes drawn, class labels displayed.

```
1: def YOLOv7():
     // Defining the model structure;
     Model ← YOLOv7();
     // Defining the optimizer;
     Optimizer ← Adam();
2: def Loss():
     // Training;
   for epoch ← 1 to num_epochs do
     // Iterate over the dataset;
     for images, labels ∈ dataset do
       // Forward propagation;
       Outputs ← model(images);
       // Calculating the loss;
       Loss ← Loss(outputs, labels);
       // Back propagation;
       Optimizer.zero_grad();
       Loss.backward();
       // Updating parameters;
       Optimizer.step();
3:   end
4: end
   // Detection;
   def ObjectDetection(image):
5:    Outputs ← model(image);
       Boxes ← NMS(outputs);
       return boxes
   // Test;
   for image ∈ test_dataset do
6:   Boxes ← ObjectDetection(image);
     // Bounding boxes drawn;
     Draw_boxes(image, boxes);
     // Class labels displayed;
     Show_labels(image, boxes);
7: end
```

The structural architecture of the improved YOLOv7 tiny network is shown in Figure 2. First, we add a 1 × 1 convolution and an identity connection to the 3 × 3 convolution, and use convolutional reparameterization [28] to boost the inference capability of the model. Secondly, in order to decrease background interference, the CA [29] module was added to the network's backbone. Then, GAM [30] module is added to the head to extract crucial information. Finally, this paper proposes focal CIOU loss, which replaces the bounding box loss CIOU loss [31] to make high-quality anchor boxes the main attention of the regression the process.

The reparameterized visual geometry group (RepVGG) network is built in the VGG style, and the general structure is similar to ResNet. The reparameterization technology is used to realize the possibility of complicated training and easy inference. By adopting a similar idea, as shown in Figure 3, we improve the 3 × 3 convolution in YOLOv7 by adding 1 × 1 convolution and identity connection between 3 × 3 convolution, so that the network can use a relatively complex structure to train and extract richer features in the occluded environment during training. After the training is completed, it is reparameterized into a relatively simple ordinary convolution for inference, which decreases a lot of model

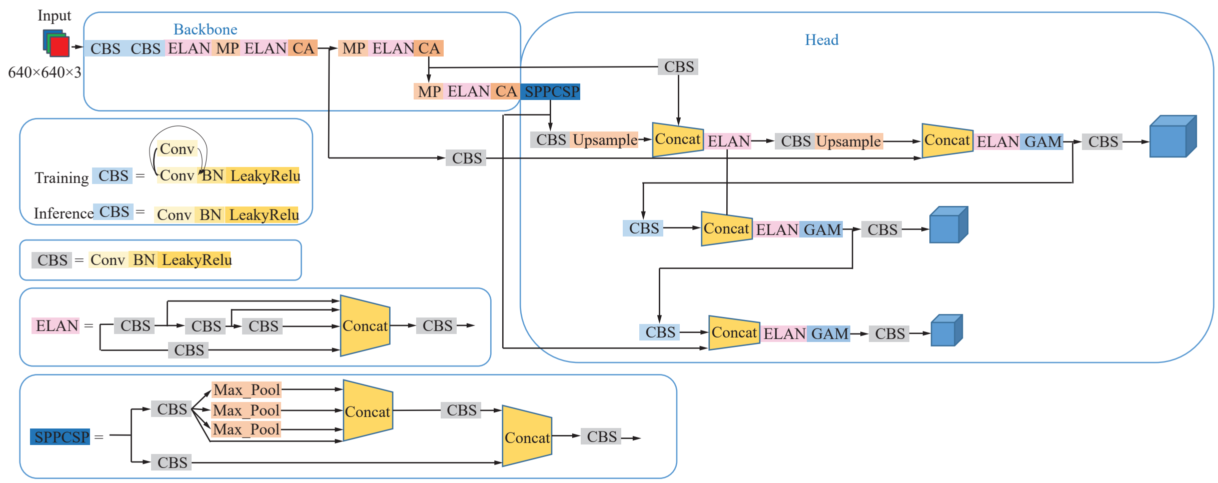

Figure 2 Diagram of improved YOLOv7 tiny network structure.

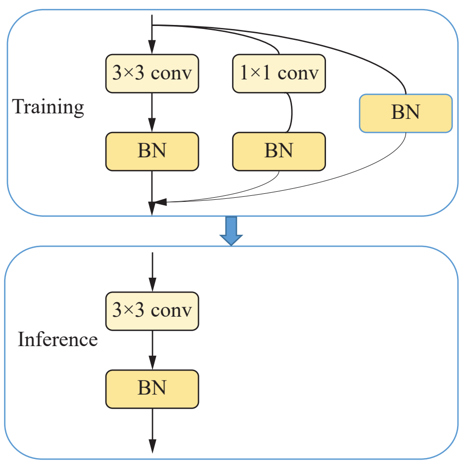

Figure 3 Diagram of improved 3 × 3 convolution.

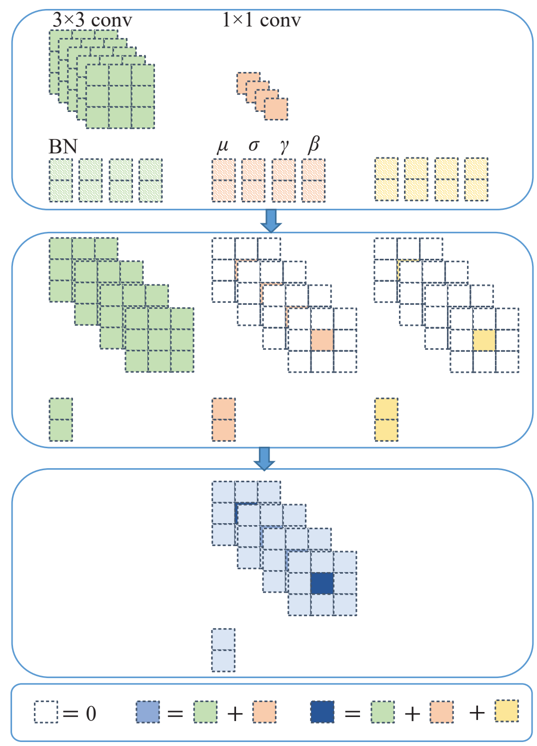

Figure 4 Diagram of the process of reparameterization.

parameters and strengthens the model's capability for inference.

As shown in Figure 4, Convolutional and batch normalization (BN) layer fusion and convolutional branch fusion are the two primary reparameterization process.

1) Convolutional and BN layer fusion

The convolutional layer is formulated as follows:

$$\mathrm{conv}(x) = w(x) + b \tag{1}$$

where x stands for the input feature map, w stands for the weight of the convolutional layer, and b stands for the bias of the convolutional layer. And the formula for BN layer is as follows:

$$\mathrm{BN}(x) = \frac{x - \mu}{\sqrt{\sigma_1^2 + \epsilon}}\gamma_1 + \beta \tag{2}$$

where μ, σ_1, γ_1, and β represent the average, variance, learning scaling factor, and bias, respectively, and ϵ is to avoid having a zero denominator. Substitute (1) into (2) as follows:

$$\mathrm{BN}(\mathrm{conv}(x)) = \frac{w(x) + b - \mu}{\sqrt{\sigma_1^2 + \epsilon}}\gamma_1 + \beta \tag{3}$$

This simplifies to

$$\mathrm{BN}(\mathrm{conv}(x)) = \frac{w(x)}{\sqrt{\sigma_1^2 + \epsilon}}\gamma_1 + \left(\frac{b - \mu}{\sqrt{\sigma_1^2 + \epsilon}}\gamma_1 + \beta\right) \tag{4}$$

Let

$$\begin{cases} W_{\mathrm{fused}} = \dfrac{w}{\sqrt{\sigma_1^2 + \epsilon}}\gamma_1 \\ B_{\mathrm{fused}} = \dfrac{b - \mu}{\sqrt{\sigma_1^2 + \epsilon}}\gamma_1 + \beta \end{cases} \tag{5}$$

The final fusion result is as follows:

$$\mathrm{BN}(\mathrm{conv}(x)) = W_{\mathrm{fused}}(x) + B_{\mathrm{fused}}(x) \tag{6}$$

where W_{fused} and B_{fused} are the weights and biases of the new convolutional layer obtained after fusing the convolutional layer and BN layer, respectively.

2) Convolutional branch fusion

As can be seen in Figure 4, the first step in the process is for the 1 × 1 convolution to be converted into 3 × 3 convolution, that is, the 1 × 1 convolution kernel is padding into 3 × 3 convolution kernel. Secondly, for the identity map, it is first converted into an 1 × 1 convolution, and then, by padding, it gets converted to a 3 × 3 convolution. Finally, three 3 × 3 convolutional branches are connected to fuse the multi-branch convolution structure into a single-branch convolution. The down-sampling convolutional layer of YOLOv7 tiny is composed of convolutional, BN layer and leaky rectified linear unit (LeakyReLu) [32]. In addition, YOLOv7 also accomplishes the fusing of the convolutional and BN layer in the inference process, allowing YOLOv7 tiny can use reparameterization.

2. Coordinate attention

Because the background of the road is complex and there is more interference information, the network must focus on the region of interest. Deep learning has made extensive use of the attention mechanism. By assigning varying weights to varying parts of the input, this enables the neural network to focus more on the main information of the input and ignore the secondary or unimportant information, so as to enhance the effectiveness and precision of the task. CA module is a small number of parameters. It enables the capture of cross-channel information as well as taking into account direction-aware and location-sensitive information to help the model more precisely detect and identify the region of interest and increase the robustness of the model to occluded objects.

Given input X, width W, height H, and channel C, as demonstrated in Figure 5, In order to get feature maps both in the height and width directions, the input feature maps are obtained by global average pooling (Avg pool) in the vertical and horizontal directions by $(1,W)$ and $(H,1)$, respectively. Specifically, the output of height and width is

$$Z^{\mathrm{h}}=\frac{1}{W}\sum_{i=0}^{W-1}X_{\mathrm{c}}(h,i) \tag{7}$$

$$Z^{\mathrm{w}}=\frac{1}{H}\sum_{j=0}^{H-1}X_{\mathrm{c}}(j,w) \tag{8}$$

where $X_{\mathrm{c}}(h,i)$ and $X_{\mathrm{c}}(j,w)$ is the value of channel C at position (h,i) and (j,w), respectively. Then, the obtained feature maps of the two directions are combined together and sent to the shared 1 × 1 convolution to reduce the quantity of channels to C/r, and then the feature map through the batch normalization (BatchNorm) layer is sent to a non-linear activation function to obtain the feature map f, where r is the reduction ratio.

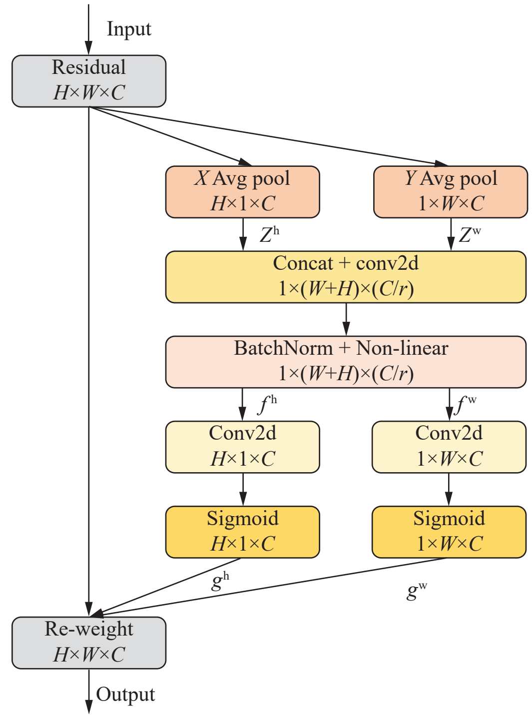

Figure 5 Structural framework of CA module.

$$f=\delta(F([Z^{\mathrm{h}},Z^{\mathrm{w}}])) \tag{9}$$

where $[\cdot,\cdot]$ represents the concatenation operation, F is a convolutional transformation, and δ is a non-linear activation function. Then, the feature map f is convolved according to the original height H and width W to obtain f^{h} with the number of channels $(C/r)\times H$ and f^{w} with the number of channels $(C/r)\times W$, respectively. After the sigmoid activation function, the feature map's attention weights g^{h} and g^{w} in height and width are got, respectively.

$$g^{\mathrm{h}}=\sigma_{\mathrm{s}}(F_{\mathrm{h}}(f^{\mathrm{h}})) \tag{10}$$

$$g^{\mathrm{w}}=\sigma_{\mathrm{s}}(F_{\mathrm{w}}(f^{\mathrm{w}})) \tag{11}$$

where σ_{s} represents the sigmoid linear activation function,which can help in changing the weights to the range of 0 to 1. F_{h} and F_{w} are two transformation functions with 1 × 1 kernels, which are used to generate two feature vectors. Finally, the original feature map is performed on the channel multiplication weighting calculations to generate the output feature map y_{c} with attention weights in the height and width directions.

$$y_{\mathrm{c}}(i,j)=x_{\mathrm{c}}(i,j)\times g^{\mathrm{h}}(i)\times g^{\mathrm{w}}(j) \tag{12}$$

We have added a CA module after layers 14, 21, and 28 in the backbone. These layers were chosen because they are what the backbone passes to the Head, thus we are applying the CA module before the output passes to the head. Without significantly affecting inference time, these three CA modules layers significantly improve the model's performance.

3. Global attention mechanism

The GAM module utilizes the sequential channal-spatial attention mechanism of the convolutional block attention module (CBAM) [33], whose submodules have been redesigned. The GAM module boosts global dimensional interactions and decreases loss of information to enhance neural network accuracy. Channel, spatial width, and spatial height are three-dimensions that can be captured as primary features so as to locate the occluded objects more accurately. The module structure is illustrated in Figure 6. Given the input feature map $\boldsymbol{F}_1$ with width W, height H, and number of channels C, the state between the intermediate feature map $\boldsymbol{F}_2$ and the output feature map $\boldsymbol{F}_3$ can be defined as follows.

$$\boldsymbol{F}_2 = \boldsymbol{M}_{\mathrm{c}}(\boldsymbol{F}_1) \otimes \boldsymbol{F}_1 \tag{13}$$

$$\boldsymbol{F}_3 = \boldsymbol{M}_{\mathrm{s}}(\boldsymbol{F}_2) \otimes \boldsymbol{F}_2 \tag{14}$$

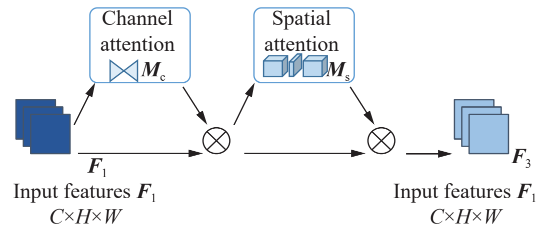

Figure 6 Diagram of the GAM module structure.

where $\boldsymbol{M}_{\mathrm{c}}$ and $\boldsymbol{M}_{\mathrm{s}}$ stand for channel attention and spatial attention feature maps, respectively; $\otimes$ means multiplication element-wise.

As shown in Figure 7, the channel attention submodule firstly performs three-dimensional transformation to retain the three-dimensional information of the input, and the feature map after dimension transformation is input to multilayer perceptron (MLP), which is a multi-layer perceptron with two layers, and then converted to the original dimension, sigmoid activation function is used for processing output.

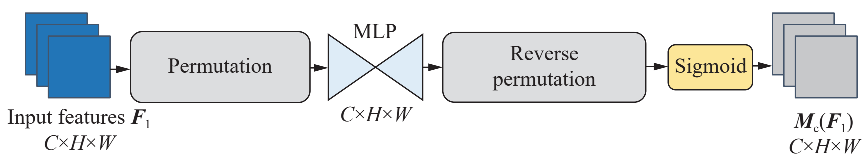

Figure 7 Diagram of channel attention submodule structure.

The spatial attention submodule mainly utilizes convolution processing. As shown in Figure 8, Firstly, a 7 × 7 convolution kernel is utilized to decrease the number of calculation by reducing the number of channels, and the $(C/r) \times H \times W$ feature map is obtained. The number of channels is further increased by a 7 × 7 convolution kernel to get a $C \times H \times W$ feature map, keeping the input and output channels consistent, and finally output by sigmoid activation function.

We add three GAM modules before sending them to the last three convolutional layers because these layers are sent directly to the last detection layer, which considerably enhances the model's capacity by capturing the three-dimensional features.

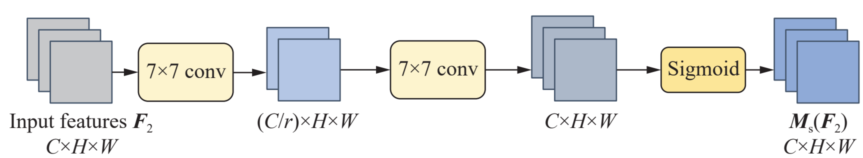

Figure 8 Diagram of spatial attention submodule structure.

4. Focal CIOU loss

Equation (15) shows the YOLOv7 loss function.

$$\mathrm{LOSS} = L_{\mathrm{boxloss}} + L_{\mathrm{confloss}} + L_{\mathrm{classloss}} \tag{15}$$

where L_{boxloss} represents regression loss, L_{confloss} represents confidence loss, and $L_{\mathrm{classloss}}$ represents classification loss. The bounding box regression loss is CIOU loss, while the confidence loss and classification loss of YOLOv7 are both binary cross-entropy losses. The overlap rate, box ratio, ground truth to expected box distance, and penalty term are all factors considered by the CIOU loss. Equation (16) shows the CIOU loss.

$$L_{\mathrm{CIOUloss}} = 1 - \left[\mathrm{IOU} - \frac{\rho^2(b^{\mathrm{p}}, b^{\mathrm{gt}})}{c_{\mathrm{d}}^2} - \alpha v\right] \tag{16}$$

$$\alpha = \frac{v}{(1 - \mathrm{IOU}) + v} \tag{17}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{\mathrm{gt}}}{h^{\mathrm{gt}}} - \arctan\frac{w}{h}\right)^2 \tag{18}$$

where $\mathrm{IOU} = |b^{\mathrm{p}} \bigcap b^{\mathrm{gt}}| / |b^{\mathrm{p}} \bigcup b^{\mathrm{gt}}|$, b^{p} represents the prediction bounding box, b^{gt} represents the ground truth; c_{d} represents the diagonal distance of the smallest closure rectangle that contains both the predicted box and the ground truth, and $\rho^2(b^{\mathrm{p}}, b^{\mathrm{gt}})$ represents the distance among the predicted box's center point and the ground truth center point; α represents the balance parameter; v can be used for measuring whether consistently the aspect ratio miantained.

The number of high-quality anchor boxes is significantly less than that of low-quality anchor boxes in bounding box regression resulting in inaccurate positioning, due to the issue of unbalanced training data under the occlusion condition. Inspired by the idea of focal Efficient Intersection over union (EIOU) [34], this paper proposes focal CIOU to improve the contribution of high-quality anchor boxes with large IOU in the model and suppress irrelevant anchor boxes. Make high-quality anchor boxes the main attention of the regression the process. To get the focal CIOU loss, we reweight the CIOU loss using the value of the IOU, which is calculated as follows:

$$L_{\mathrm{focalCIOU}} = \mathrm{IOU}^{\gamma} L_{\mathrm{CIOU}} \tag{19}$$

The γ inhibition is to control the abnormal value of parameters. We use $\gamma = 0.5$ as set in focal EIOU and take it as the default value for our experiments.

IV. Experiments

1. Experimental setup

We perform network model evaluation on the widely used object detection benchmark datasets pattern analysis, statistical modelling and computational learning visual object classes (PASCAL VOC) 2007 [35] and PASCAL VOC 2012 [36], trained with VOC 07+12 (VOC 2007 train, trainval and VOC 2012 train, trainval). with a total of 16551 images. We used VOC07 test for testing with 4952 images. The dataset contains a variety of categories such as vehicles and pedestrians, and the picture scene is more diversified, which is more suitable for vehicle and pedestrian detection under occluded conditions. We select to train from scratch in order to confirm the accuracy of the improved experiments, with batch and epoch set to 32 and 300, respectively. A Nvidia GeForce RTX4090 24 G graphics card was utilized for training. All models are implemented in PyTorch and trained on Ubuntu20.04 system with CUDA12.1 and Python 3.9.

2. Evaluation metrics of the algorithm

So as to measure the effectiveness of the proposed improved YOLOv7 tiny algorithm, Precision, Recall, mean average Precision (mAP), and F1-score are used as measurement indicators.

Precision represents the proportion of true predictions out of all predictions. The formula is as follows:

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \times 100\% \tag{20}$$

where TP stands for "true positive samples" and represents the quantity of things that were correctly detected; FP stands for "false positive samples", indicates the quantity of objects that were mistakenly detected.

Recall represents the proportion of true predictions out of all ground truths. The formula is as follows:

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \times 100\% \tag{21}$$

where FN stands for "false negative samples", indicates the quantity of unnoticed or missed detections.

The average of all class average precision (AP) is known as mAP, which is used to measure the overall detection accuracy of the object detection algorithm. The formula is as follows:

$$\mathrm{mAP} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{AP}_i \tag{22}$$

The F1-score represents the harmonic average of recall and precision; a higher F1-score indicates greater object detection precision. The formula is as follows:

$$\text{F1-score} = 2 \times \frac{\mathrm{Percision} \times \mathrm{Recall}}{\mathrm{Percision} + \mathrm{Recall}} \tag{23}$$

3. Experimental results

On the PASCAL VOC dataset, the proposed improved YOLOv7 tiny algorithm's detection capability is experimentally evaluated. We provide an image of several loss function variations, including regression loss, confidence loss, and classification loss. The average of the focal CIOU loss function is known as the bounding box loss, where smaller values indicate higher accuracy. The loss value shows a steady decrease to reach convergence as the quantity of iterations rises, shown in Figure 9.

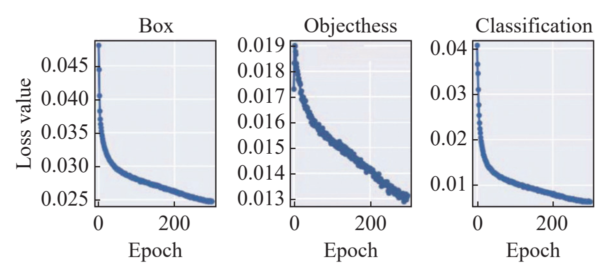

Figure 9 Change of loss function.

So as to further verify the superiority of the proposed model, we utilize several different object detection models trained on the same dataset to guarantee the consistency of training parameters and training environments, including YOLOv7 tiny, YOLOv8n, YOLOv5n, faster R-CNN, mask R-CNN, SSD. By comparing the results on PASCAL VOC dataset, as shown in Table 2, the improved YOLOv7 algorithm is superior to other detection algorithms. The map is 2.8% higher than YOLOv7 tiny. It is 3.2%, 6.7%, 4.7%, 3.3%, and 4.0% higher than YOLOv8n, YOLOv5n, faster R-CNN, mask R-CNN, and SSD, respectively. Meanwhile, our method

Table 2 Performance comparison of object detection models on PASCAL VOC dataset

Method	Precision (%)	Recall (%)	F1-score (%)	Map @ 0.5 (%)	Map @ 0.5:0.95 (%)	FPS
SSD300	76.8	73.2	73	78.2	48.8	59
SSD512	79.3	73.3	76	79.8	55.2	22
Faster R-CNN	81.0	75.4	76	77.5	52.3	17
Mask R-CNN	81.5	76.2	77	78.9	58.2	15
YOLOv5n	75.2	69.5	70	75.5	49.6	126
YOLOv8n	79.7	71.3	75	79.0	58.9	80
YOLOv7 tiny	77.2	73.9	76	79.4	54.4	275
Ours	78.2	76.9	77	82.2	59.6	252

achieves 252 frames per second (FPS), which is comparable to YOLOv7 tiny. However, mask R-CNN only has 15 FPS and faster R-CNN only has 17 FPS, and our method is 16.8% and 15% higher than mask R-CNN and faster R-CNN, respectively. YOLOv7 tiny has the same accuracy as mask R-CNN, but it is 16.8% FPS higher. Experimental results show the effectiveness of our work.

We selected three sets of pictures in different scenes to show the results on PASCAL VOC dataset. The selected pictures have complex backgrounds, different degrees of occlusion. As shown in Figure 10, YOLOv7 tiny only detected five of the seven cars in the first image, resulting in missed detection, while our method could detect all the targets. As can be seen from the second and third figures in Figure 10, for occluded vehicles and pedestrians, our method can still accurately detect vehicles and pedestrians compared with YOLOv7 tiny.

4. Ablation experiments

To evaluate the effectiveness of different refinements on model performance in an autonomous driving environment, we conducted ablation experiments. We first train on the PASCAL VOC dataset using YOLOv7 tiny. Then, based on YOLOv7 tiny, model reparameterization

(a) (b)

Figure 10 Vehicle and pedestrian detection under occlusion. (a) Before improvement; (b) After improvement.

is added, and then CA and GAM modules are introduced. Finally, the bounding box loss uses FCIOU instead of CIOU. Table 3 shows the experimental results. When compared to prior improvements, the accuracy of bicycle is increased by 4.5%, the accuracy of bus is increased by 2.6%, the accuracy of car is increased by 1.9%, the accuracy of motorbike is increased by 2.2%, and the accuracy of person is increased by 1.5%. The results of the experiment demonstrate that our improvement can increase the precision of the model. And successfully detect occluded objects of vehicles and pedestrians.

Table 3 Ablation experiments on PASCAL VOC dataset

Method	Accuracy (%)					
	Bicycle	Bus	Car	Motorbike	Person	Map
YOLOv7	87.7	87.4	91.2	89.4	87.1	79.4
YOLOv7+Rep	89.4	88.5	91.6	88.7	87.4	80.0
YOLOv7+Rep+CA	89.8	89.0	91.5	89.1	87.8	80.6
YOLOv7+Rep+CA+GAM	90.2	89.1	92.2	89.2	88.3	81.5
YOLOv7+Rep+CA+GAM+FCIOU	92.2	90.0	93.1	91.6	88.6	82.2

In order to verify the effectiveness of the algorithm, we respectively use CBAM, CA and GAM as attention mechanism, add them to the YOLOv7 tiny algorithm, and test them on the PASCAL VOC dataset. As shown in Table 4. Compared with the original YOLOv7, the accuracy of CBAM-YOLOv7 is reduced by 1.3%. CA-YOLOv7 has a 1% improvement, and GAM-YOLOv7 has a 1.2% improvement. From the comparative analysis of the experimental results, it can be concluded that higher accuracy can be obtained by using the CA module and the GAM module. The performance of the algorithm has improved.

Table 4 Comparison of different attention

Method	Accuracy (%)					
	Bicycle	Bus	Car	Motorbike	Person	Map
YOLOv7	87.7	87.4	91.2	89.4	87.1	79.4
YOLOv7+CBAM	88.3	87.0	90.5	87.5	86.1	78.1
YOLOv7+CA	89.2	88.5	91.5	89.0	87.6	80.4
YOLOv7+GAM	88.0	87.8	86.0	89.7	88.2	80.6

We also verified the results of different loss functions. As shown in Table 5. Compared with CIOU in the original YOLOv7, although the use of EIOU improves in bicycles and cars, the overall accuracy decreases by 1.8%, and focal EIOU decreases by 0.9%. The improvement is 1.1% using the proposed focal CIOU.

Table 5 Comparison of different loss functions

Method	Accuracy (%)					
	Bicycle	Bus	Car	Motorbike	Person	Map
CIOU	87.7	87.4	91.2	89.4	87.1	79.4
EIOU	88.1	86.2	92.4	81.8	80.4	77.6
Focal EIOU	87.8	86.4	90.5	89.0	88.2	78.5
Focal CIOU	89.2	88.6	92.1	90.5	88.3	80.5

V. Conclusions

In this study, we design an improved YOLOv7 tiny algorithm for vehicle and pedestrian detection under occlusion in autonomous driving. To attain this target, the 1×1 convolution and identity connection are added to the 3×3 convolution in the network, which can enhance the model's memory utilization and inference speed, and improve the detection accuracy. Then, the coordinate attention and global attention mechanism were used to focus the model concentration on the region of interest and increase the detection performance of the algorithm. Finally, due to the imbalanced training samples of CIOU loss, focal CIOU loss is proposed to enhance the overall performance of the algorithm. This paper uses the PASCAL VOC dataset for experiments, and the results of the experiments demonstrate that our method can strengthen the model's robustness for object detection in the presence of occlusion and blur. However, the detection of vehicles with tiny objects in occlusion situations continues to be a significant challenge in the field of object detection. We will keep researching ways to boost object detection's precision in the future to give autonomous driving a stronger auxiliary function.

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